



SOCIETY OF ELECTRONICS ENGINEERS

INDIAN INSTITUTE OF TECHNOLOGY (ISM), DHANBAD



presents

INTRODUCTION TO

THE DEPARTMENT OF ELECTRONICS ENGINEERING

Introducing freshers to the
department, faculties and academics .



Quick Poll: Why ECE?

 I CHOSE ECE



Raise your hand if Electronics & Communication Engineering was your intentional top preference.

 JEE BROUGHT ME HERE



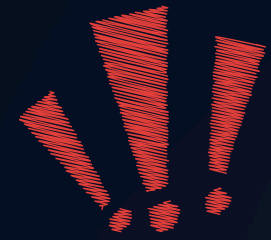
Raise your hand if your rank/seat allocation made the decision for you.



Scan the QR code for a quick live QUIZ !!



Answer fast and climb the LEADER BOARD



ECE is bigger than just a breadboard.

CORE DOMAINS:

VLSI
Silicon & chips



Embedded
Hardware +
software



Communication

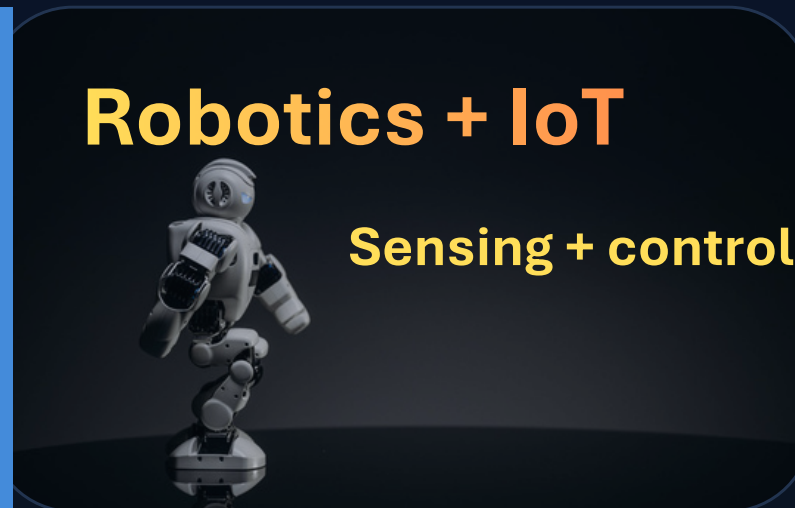


Signal
Processing
Audio, image,
video



Robotics + IoT

Sensing + control



AI Hardware

AI at the
edge





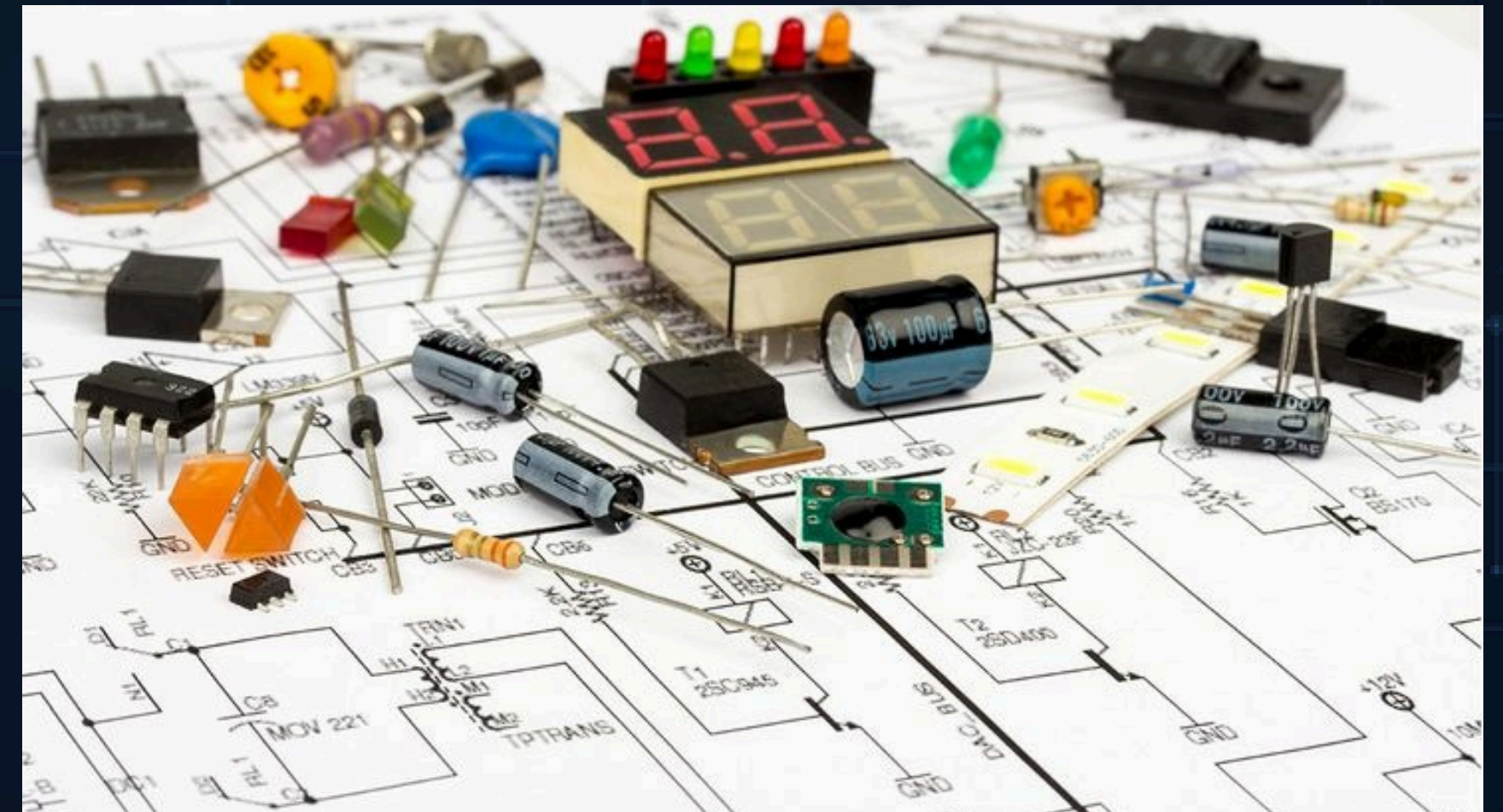
ANALOG ELECTRONICS

WHAT YOU LEARN

- Diodes, BJTs & MOSFETs
- Operational Amplifiers & Amplifiers
- Oscillators & Filters
- Analog Circuit Design

WHERE IT SHOWS UP

- Audio & Communication Systems
- Sensor Interfaces
- Biomedical Instruments
- Signal Conditioning & Electronics





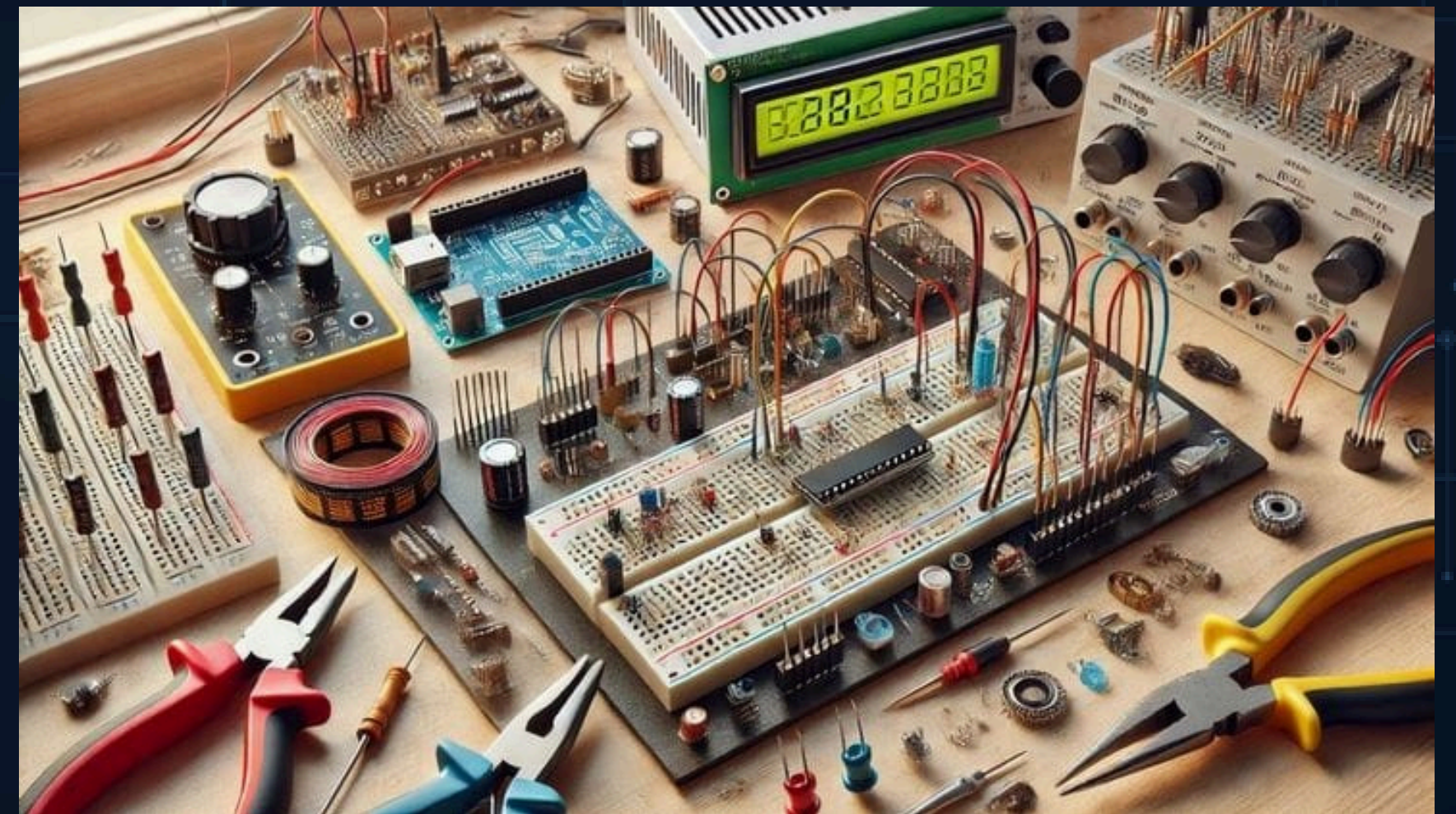
DIGITAL ELECTRONICS

WHAT YOU LEARN

- Boolean Algebra & Logic Gates
- Combinational & Sequential Circuits
- Flip-Flops, Counters & Registers
- Memory & Finite-State Machines

WHERE IT SHOWS UP

- FPGA & VLSI Design
- Verilog / SystemVerilog
- Processor & Computer Architecture
- Digital System Design





VLSI AND SEMICONDUCTOR DESIGN

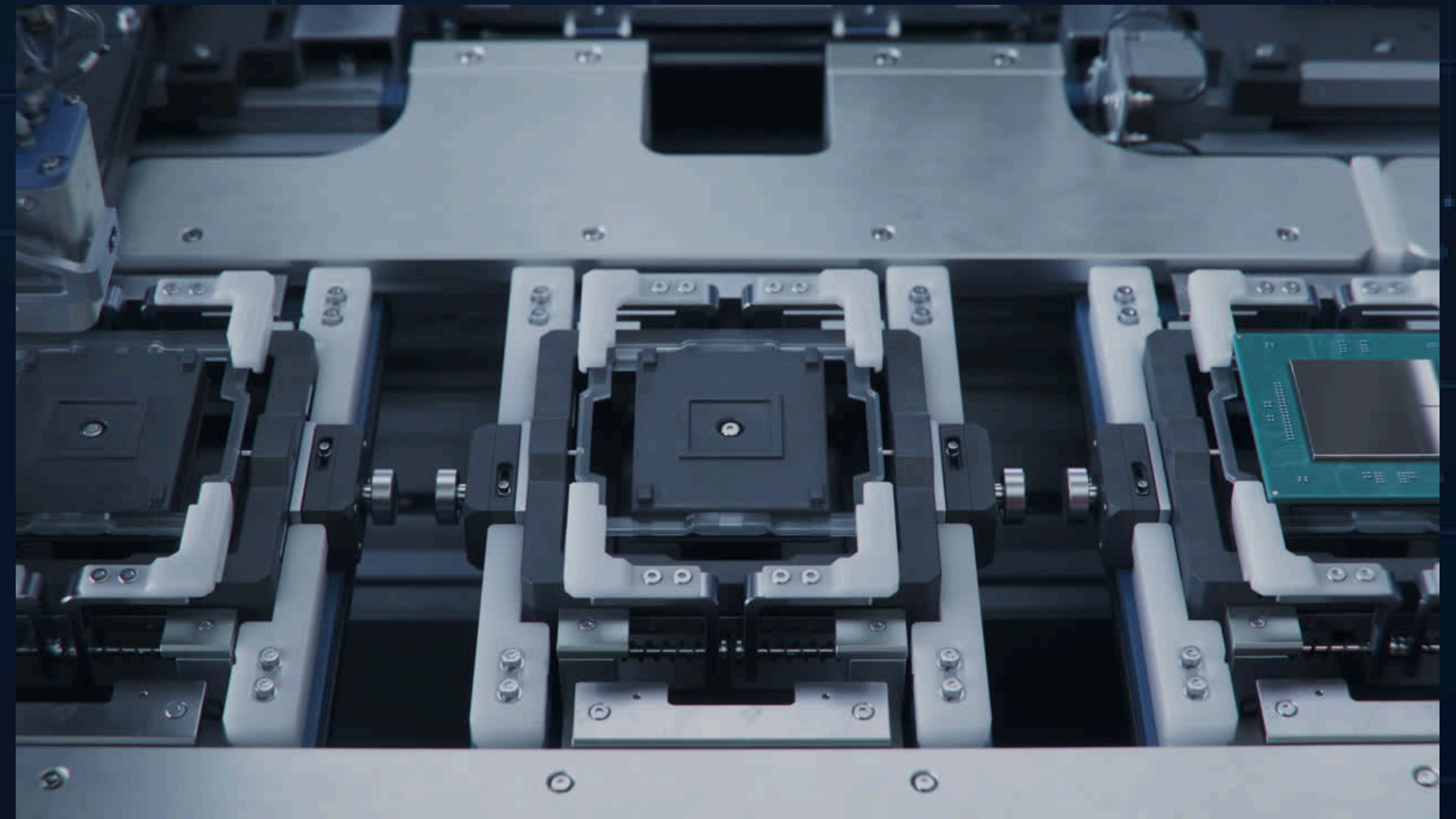


WHAT YOU LEARN

- Digital System Design
- CMOS technology
- Verilog / VHDL
- Physical synthesis

WHERE IT SHOWS UP

- FPGA & VLSI Design
- Verilog / SystemVerilog
- Processor & Computer Architecture
- Digital System Design





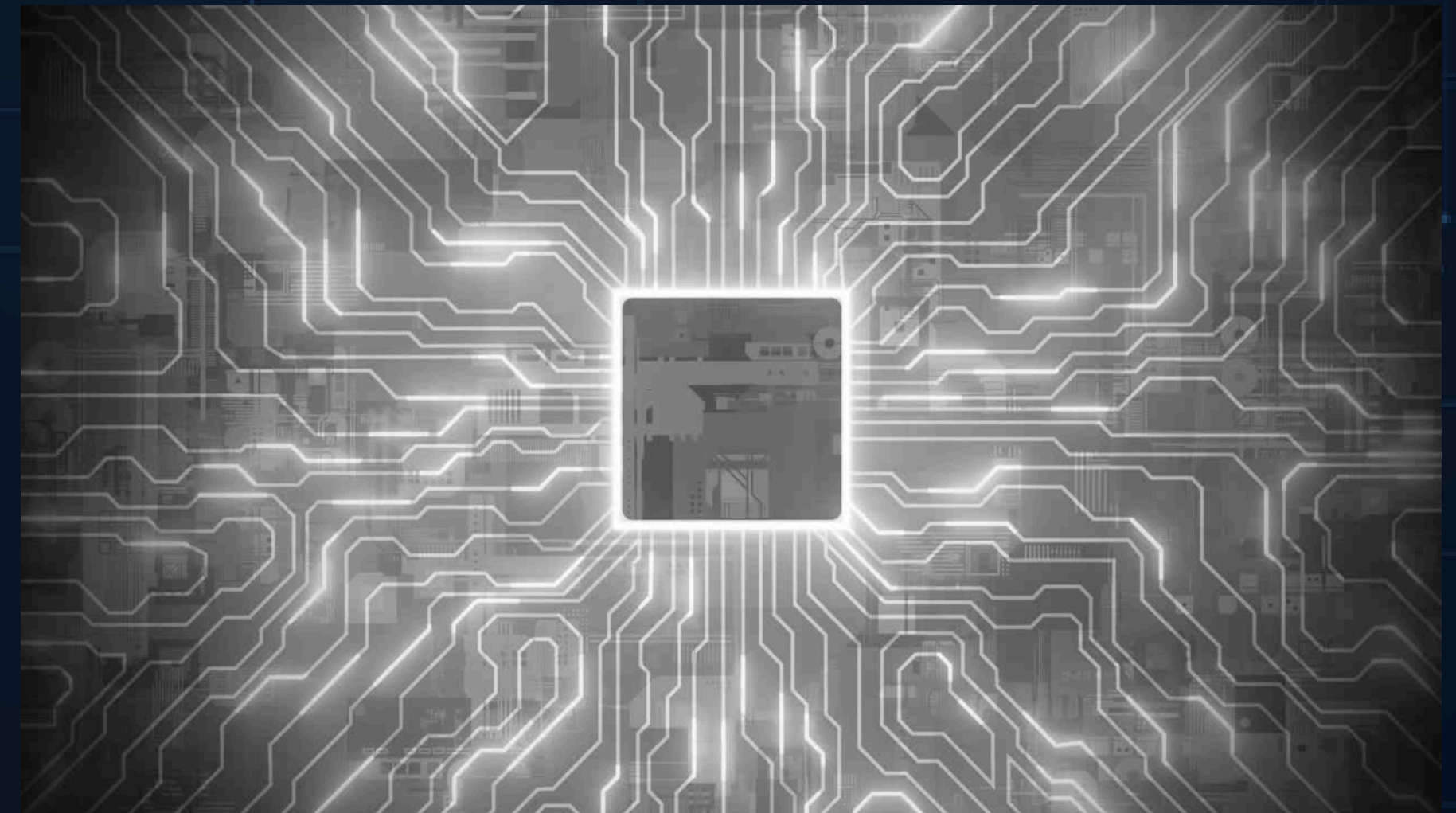
EMBEDDED AND IoT

WHAT YOU LEARN

- C / C++
- RTOS
- ARM / STM32 / ESP32
- Sensors + Actuators

WHERE IT SHOWS UP

- Examples: projector
- Smartwatch
- ABS
- Drones
- Pacemakers





COMMUNICATION SYSTEMS

WHAT YOU LEARN

- Digital Communication & Modulation
- Wireless & RF Systems
- Signal Processing & Filtering
- Wave Propagation & Networking

WHERE IT SHOWS UP

- 4G / 5G / 6G & Wi-Fi
- Satellite & Optical Communication
- Voice & Speech Recognition
- Radar, Medical Imaging & Wearables





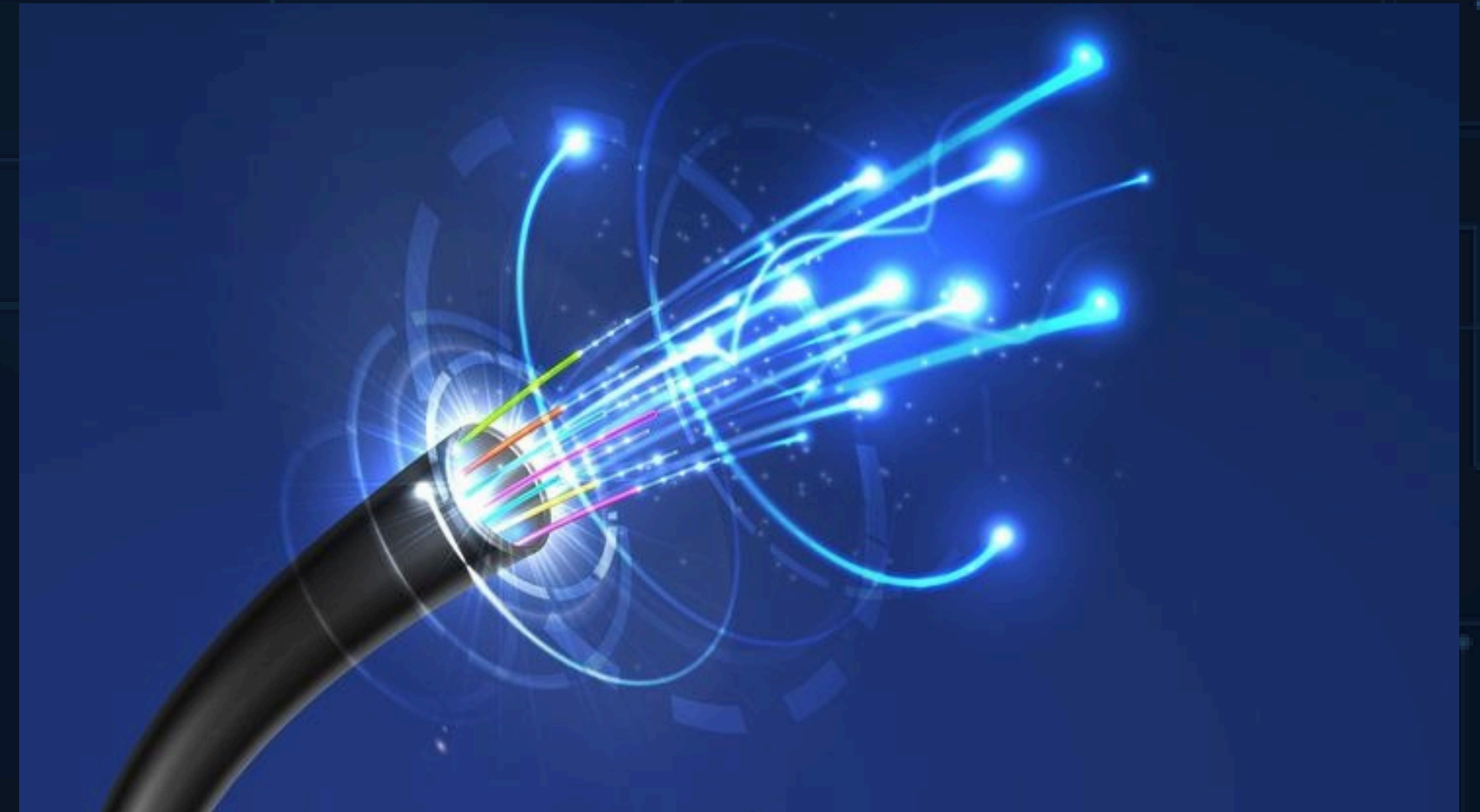
OPTICAL SYSTEMS

WHAT YOU LEARN

- Geometrical & Wave Optics
- Lasers & Optical Fibers
- Photodetectors & Optical Devices
- Light–Matter Interaction

WHERE IT SHOWS UP

- Fiber-Optic Communication
- Laser Technology
- Cameras & Optical Sensors
- Medical Imaging & Biomedical Devices





AI AND QUANTUM HARDWARE



WHAT YOU LEARN

- Parallel hardware architectures
- AI acceleration
- Low-latency computing
- Edge deployment

WHERE IT SHOWS UP

- Autonomous vehicles
- surgical robots
- on-device intelligence





Where can ECE take you?

CORE HARDWARE + VLSI

ASIC • RTL • Analog •
Physical Design

TI • NXP • Qualcomm
• Intel • NVIDIA •
AMD • MediaTek

EMBEDDED + FIRMWARE

Firmware • RTOS •
Systems Integration

Bosch • Samsung
Semiconductor •
NXP • Qualcomm

COMMUNICATION + DSP

RF • Wireless • Signal
Processing

Qualcomm • Apple •
Ericsson • Samsung
R&D • Telecom •
Defense

SOFTWARE • AI • DATA

SDE • ML Engineer •
Quant Analyst

Google • Microsoft •
Amazon • Oracle •
HFT firms

The branch is a launchpad—not a single job title.



OPPORTUNITIES

Women in Technology: Mentorship pathways

NXP SEMICONDUCTORS

From the 3rd semester: mentorship, technical workshops, tool exposure and potential internship consideration.

WiSH • TEXAS INSTRUMENTS

For 4th-semester students: technical bootcamps, hardware design training, mentorship and exposure to silicon design.



SPECTRUM

THE ANNUAL ELECTRONICS TECHNICAL FEST

DOMAINS IN SPECTRUM '26

- Hardware and Circuit Design
- VLSI and Digital Logic
- Embedded Systems and Robotics
- Signal Processing and Data Analysis
- Software and AI

+ 2 Movie Nights (of your choice)





SPECTRUM

THE ANNUAL ELECTRONICS TECHNICAL FEST





UPCOMING FUN AND INTERACTIVE EVENTS



ICE BREAKER



FRESHERS PARTY



SILICON TALKS



ONGOING RESEARCH IN THE DEPARTMENT



- Development of lab-scale quantum fiber network, and design of plug-and-play photonic integrated modules for its transmitter & receiver towards secure 6G communication
- Development of Quantum Algorithms for Next Generation Wireless Communication Systems
- Silicon Photonic Chip based $N \times N$ Photonic Switches for 6G and Beyond
- NeTS Small NSF-DST Modernizing Underground Mining Operations with Millimeter-Wave Imaging and Networking
- Capacity building for human resource development in Unmanned Aircraft System (Drone and related Technology)
- Ultra-Low Power Neuromorphic Spiking Architecture for Assistive Smart Glass
- Bandwidth Adaptive RF Power Amplifier Modeling for Cross Bandwidth Testing



ONGOING RESEARCH IN THE DEPARTMENT



- **TeraHertz Communication: Prototype Development and Algorithm Design**
- **Development of planar microwave sensor for adulteration detection**
- **Demonstration of Graphene FET Fabricated on hBN/SiO₂/Si Stack and Exploration of Role of Defects on Device Characteristic**
- **Development of 2D Material-Enhanced Silicon Photonic Devices for High-Performance Point-to-Point Optical Communication**
- **Harnessing the Harmonics in Sequential Load Modulated Balanced Amplifiers for Efficiency and Linearity Enhancement for 5G/6G Applications**
- **Memristor based Multilayer Neural Networks (MNN) and its application in Neuromorphic System**



ONGOING RESEARCH IN THE DEPARTMENT



- Development of ceramic dielectric material potentially viable for microwave devices
- Critical Analysis, Design, EM Modelling & Optimisation of millimetre wave planar antenna
- ChalPV: Chalcogenide-Perovskite (BaZrS₃) based Photovoltaic Devices
- Design and Development of Substrate Integrated Waveguide based Self-multiplexing Antennas for 5G communication systems
- Design and Development of Frequency Stable and High Q-Factor Optoelectronic Oscillator Assisted by Microwave Photonic Techniques
- Design of a webserver-based hybrid physiological sensor with optical cloth for real-time health specialist care
- Smart Radio Environments: Implementation and Deployment for Targeted Use Case



ONGOING RESEARCH IN THE DEPARTMENT



- **Generation of high chirp rate dual chirp microwave waveform in Ku band using novel microwave photonic techniques for high performance radar application**
- **Multi-Carrier Waveforms for intelligent Reflecting Surface (IRS) assisted Next-Generation Wireless Communication Systems**
- **Development of Integrated Photonic Devices Incorporating Silicon Plus Photonics for Next-Generation Optical Communication**



SEMICON INDIA

NATIONAL RECOGNITION — SEMICON INDIA

PROJECT & RECOGNITION

- APEEC1 IC
- Low-Power Analog Memristor Emulator IC
- Developed by Electronics Engineering, IIT (ISM) Dhanbad
- Recognized at SEMICON India 2025
- National-level recognition

RESEARCH & IMPACT

- Low-power analog circuit design
- Memristor-based circuit technology
- Focus on next-generation electronic systems
- Showcases innovation in semiconductor research
- Guided by Prof. Rajeev Kumar Ranjan





PMRF AND THE RESEARCH JOURNEY

The Prime Minister's Research Fellowship (PMRF) is a prestigious national initiative that supports and encourages students pursuing advanced research in India.

WHAT IS PMRF?

- Prime Minister's Research Fellowship
- Prestigious national fellowship for doctoral research
- Supports students pursuing advanced research in India
- Encourages innovation and high-impact research

STARTING YOUR RESEARCH JOURNEY

- Build strong fundamentals during your undergraduate years
- Interact with professors and explore their research
- Participate in projects and research opportunities
- Learn relevant technical tools & skills
- Develop a mindset of curiosity, experimentation & innovation



OUR DEPARTMENT IN NAC





LABS

Main Lab 1 - NAC 2nd Floor Room no 231

Main Lab 2 - NAC 2nd Floor Room no 225

VLSI Lab - NAC 2nd Floor Room no 202

Hardware Lab - NAC 6th Floor Room no 635

Communication System Lab - NAC 2nd Floor Room no 215

Digital Signal Processing Lab - NAC 2nd Floor Room no 208

Microwave and Antenna Lab - NAC 2nd Floor Room no 232



KNOWING OUR PROFESSORS



Head of Department
Prof. Subindu Kumar
Professor
subindu@iitism.ac.in
+91-326-223-5732



Prof. Jitendra Kumar
Professor
jitendra@iitism.ac.in
+91-326-223-5402



Prof. Debjani Mitra
Professor
debjanim@iitism.ac.in



Prof. Santanu Dwari
Professor
santanu@iitism.ac.in



Prof. Sanjeev Kumar
Raghuwanshi
Professor
sanjeevrus77@iitism.ac.in



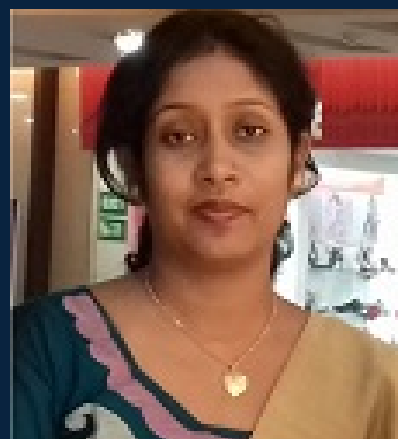
Prof. Ravi Kumar Gangwar
Professor
ravi@iitism.ac.in
+91-326-223-5903



Prof. Sushrut Das
Professor
sushrut@iitism.ac.in
+91-326-223-5496



Prof. Mrinal Sen
Associate Professor
mrinalsen@iitism.ac.in
+91-326-223-5657



Prof. Nirupama Mandal
Associate Professor
nirupama@iitism.ac.in
+91-326-223-5106



Prof. Rajeev Kumar Ranjan
Associate Professor
rajeev@iitism.ac.in
+91-326-223-5658



Prof. Amitesh Kumar
Associate Professor
amitesh@iitism.ac.in
+91-326-223-5759



Prof. Devendra Chack
Associate Professor
devendra.iitism.ac.in
+91-326-223-5660



Prof. Kaushik Mazumdar
Associate Professor
kaushik@iitism.ac.in



Prof. Manodipan Sahoo
Associate Professor
manodipan@iitism.ac.in
+91-326-223-5146



**Prof. Himanshu Bhusan
Mishra**
Associate Professor
himanshu@iitism.ac.in
+91-326-223-5016



Prof. Preety Priya
Assistant Professor
preetypriya@iitism.ac.in



Prof. Shalu Rani
Assistant Professor
shalu@iitism.ac.in



Prof. Rahul Bhattacharya
Assistant Professor
rahul@iitism.ac.in
+91-326-223-5129



Prof. Govind Murmu
Assistant Professor
govind@iitism.ac.in
+91-326-223-5123



Prof. Jaisingh Thangaraj
Associate Professor
jaisingh@iitism.ac.in
+91-326-223-5016



Prof. Swati Rajput
Assistant Professor
swatirajput@iitism.ac.in



Prof. Y. M. Asha Latha
Assistant Professor
ymashalatha@iitism.ac.in



Prof. Ugrasen Singh
Assistant Professor
ugrasen@iitism.ac.in
+91-326-223-209



Prof. Aman Sikri
Assistant Professor
amansikri@iitism.ac.in



Prof. Naresh Sharma
Assistant Professor
nareshsharma@iitism.ac.in
+91-326-223-5540



Prof. Manoj Kumar
Assistant Professor
manojkr@iitism.ac.in
+913262235337



Prof. Jeevesh Kumar
Assistant Professor
jeevesh@iitism.ac.in
+91-326-223-5457



**Prof. Prashant Kumar
Varshney**
Assistant Professor
pkvarshney@iitism.ac.in
+91-326-223-5383



Prof. Dheeraj Bharti
Assistant Professor
dheerajbharti@iitism.ac.in



Prof. Nirjhar Kumar
Assistant Professor
nirjhar@iitism.ac.in



Prof. Subhadeep Mondal
Assistant Professor
subhadeep@iitism.ac.in



**Prof. Samrat
Mukhopadhyay**
Assistant Professor
samrat@iitism.ac.in
+91-326-223-5388



COURSE STRUCTURE

3rd SEMESTER

- Electronic Devices
- Measurements and Instrumentation
- Signals and Networks
- Signals and Networks Lab
- Microprocessor and Microcontroller
- Microprocessor and Microcontroller Lab
- Hardware Description Language

4th SEMESTER

- Analog Circuits
- Electromagnetic Theory
- Electronic Devices Lab
- Principles of Communication System
- Communication System Lab
- Circuit Simulation With SPICE



COURSE STRUCTURE

5th SEMESTER

- Digital Communication
- Digital Communication Lab
- Microwave Engineering
- Microwave Engineering Lab
- Optical Engineering
- Optical Engineering Lab

6th SEMESTER

- Digital Signal Processing
- Digital Signal Processing Lab
- Control Systems
- VLSI Technology and Design
- VLSI Design Lab



COURSE STRUCTURE

7th SEMESTER

- Industrial Instrumentation
- Industrial Instrumentation lab
- Project I
- Internship

8th SEMESTER

- Machine Learning and AI
- Wireless Communication
- Wireless Communication Lab
- Project II



LEARN AND GROW

ACADEMICS & CONSISTENCY

- Attend classes regularly
- Build strong fundamentals
- Make useful notes & revise regularly
- Use tutorials and labs effectively
- Clear doubts early
- Consistency > Last-minute studying

BALANCE & COLLEGE LIFE

- Enjoy clubs, events, fests & competitions
- Balance academics with experiences
- Develop technical & communication skills
- Work on projects and explore opportunities
- Build a strong CGPA + skills + experiences
- Make the most of your college journey



PROF. DEBJANI MITRA

RESEARCH INTERESTS

- Wireless Communication
- 5G Technologies
- Communication Networks
- Machine Learning
- Statistical Signal Processing

PROFESSIONAL HIGHLIGHTS

- Co-PI in a quantum-fiber/photonics project targeting secure 6G communication.
- robotic sensor-fusion strategies using neural networks and spatial optimization
- 45 journal/conference publications

45+ Publications | 500+ Citations



Designation: Professor
Department: Electronics Engineering
Email: debjanim@iitism.ac.in
Contact Number: 9430132055



WELCOME TO ECE

You didn't just choose a branch.
You entered a field that connects the physical and digital worlds.

THANK YOU



That's the end